

Section 4.4 (cont.)

I. Infinite Limits

a. $\lim_{x \rightarrow c} f(x) = +\infty$ ($\lim_{x \rightarrow c^+} f(x) = +\infty$, $\lim_{x \rightarrow c^-} f(x) = +\infty$)

b. $\lim_{x \rightarrow c} f(x) = -\infty$ ($\lim_{x \rightarrow c^+} f(x) = -\infty$, $\lim_{x \rightarrow c^-} f(x) = -\infty$)

Examples

$$f(x) = \frac{x+1}{x-2}, \quad c = 2$$

$$f(x) = \frac{x^2 + x}{x^2 - 2x + 1}, \quad c = 1$$

$$f(x) = (x^2 + 1) \tan x, \quad x = \frac{\pi}{2}$$

$$f(x) = \ln x, \quad c = 0$$

II. Vertical Asymptote

III. Sketching the Graph of a Function $y = f(x)$ with (potentially) Asymptotes

- a. Identify the domain
- b. Identify any intercepts
- c. Identify any asymptotes
 1. Horizontal asymptotes ($y = L$ where $L = \lim_{x \rightarrow \pm\infty} f(x)$)
 2. Vertical asymptotes ($x = c$ where $\lim_{x \rightarrow c^\pm} f(x) = \pm\infty$)
- d. Identify the critical numbers: $f'(x)$ vanishes
- e. Identify the intervals of monotonicity:
 1. Intervals where $f'(x) > 0 \Leftrightarrow$ Intervals where f is strictly increasing
 2. Intervals where $f'(x) < 0 \Leftrightarrow$ Intervals where f is strictly decreasing
- f. Classify each critical point as a relative maximum, relative minimum or neither

- g. Identify the second-order critical numbers: $f''(x)$ vanishes
- h. Identify the intervals of concavity
 - a. Intervals where $f''(x) > 0 \Leftrightarrow$ Intervals where f is concave up
 - b. Intervals where $f''(x) < 0 \Leftrightarrow$ Intervals where f is concave down
- i. Identify any inflection points
- j. Construct a table with a few specific values of the function
- k. Sketch the graph of $y = f(x)$ incorporating all of the above information